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RESEARCH

Integrating Choice and Predictive Theories: A Comparative Study of Set Generation and Machine Learning in Consideration Set Modeling

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Abstract

This study introduces a comparative framework designed to evaluate the predictive capabilities of two distinct modeling approaches: traditional consideration and choice set generation models, and Machine Learning (ML) models. The framework focuses on a critical and often overlooked stage in decision–making process, which includes the set of alternatives that were considered prior to making a choice. To demonstrate the framework's applicability, a synthetic route choice scenario was employed as a case study. In this context, routes with randomly assigned attributes were generated alongside synthetic populations of decision-makers, whose socioeconomic profiles were sampled from observed survey data. Each agent was assigned a set generation models to construct their individual consideration sets. ML models were then trained on these consideration sets to predict the considered alternatives of independent decision-maker populations. Two population scenarios were examined: one homogeneous, where all agents used the same set generation model, and one heterogeneous, where models were randomly assigned across agents. In both cases, the accuracy of the prediction was evaluated, along with a comparison between existing set generation and predictive models. The results showed that certain ML models can achieve over 80% accuracy in predicting individual considered items and over 60% accuracy in predicting entire consideration sets, across both population scenarios.

Keywords Consideration set · Set generation model · Machine Learning · Prediction · Route choice

Introduction

Understanding the decision-making process is crucial for accurately predicting choices made by individuals. The objective of this work was to evaluate the predictive performance of Machine Learning (ML) models in comparison to traditional set generation methods, particularly focusing on their role in the decision-making process, specifically in determining the set of alternatives considered before the final decision. All the available options a decision-maker can potentially obtain are defined as the universal set. Alternatives that the individual is aware of, options that are within the individual's environment and pass the screening process, can be considered the awareness set. The inert set, also included in the awareness set, contains the non-preferred alternatives for that individual. The consideration set includes alternatives for which the individual will try to gather further information before the choice process. The choice set follows, comprising the few final alternatives that

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are almost equally considered (when such alternatives exist), before moving to the actual choice.

A decision-making process overview is presented in Fig. 1, emphasizing the feedback loop from the choice back to the awareness set, highlighting how past experiences can influence (even indirectly, as the dashed line depicts) future decisions (Caplin et al. 2019). Accurately predicting the consideration set, or even earlier stages (e.g., awareness set), can provide insights into latent attributes affecting decision-making. External influences and the attributes of both alternatives and decision-makers significantly shape these sets and, consequently, the final choice. Identifying the effects of consideration on decision-making can help avoid biased choice model parameters (Başar and Bhat 2004). Decision-makers typically do not evaluate all feasible alternatives; instead, they tend to restrict their consideration to attractive options based on their constraints, preferences, and experiences. Often, considered choice sets are not observed (Hoogendoorn-Lanser 2005; Prato 2009), although such observations can offer valuable insights.

This study introduces a comparative framework between choice and consideration set generation models, and ML predictive models. The framework was applied to a synthetic route choice problem, where synthetic populations of decision-makers/agents with randomly assigned socioeconomic data, sampled from observed survey data, were assigned on constructing their consideration and choice sets, from a set of routes, based on the routes' attributes and their coefficients towards them. The following existing set generation models were considered to construct each agent's consideration set: deterministic utility, random utility, Latent Class, Multinomial Logit (MNL), Generalized Logit (GenL), Dogit (Logit Captivity), Path-Size Logit (PSL), Parametrized Logit Captivity (PLC). The following ML models were then trained on these consideration sets, to later predict the considered alternatives of agents from different populations: Random Forest (RF), Decision Trees, Binary Relevance (BR), Classifier Chains (CC), Label Powerset (LP)).

The motivation for the study arises from the need to assess the feasibility of enhancing predictive capabilities and understanding agent behavior, as well as the desire to expand the analytical toolkit for consideration set modeling. This is achieved by integrating traditional choice modeling techniques with ML methods to leverage ML's ability to handle

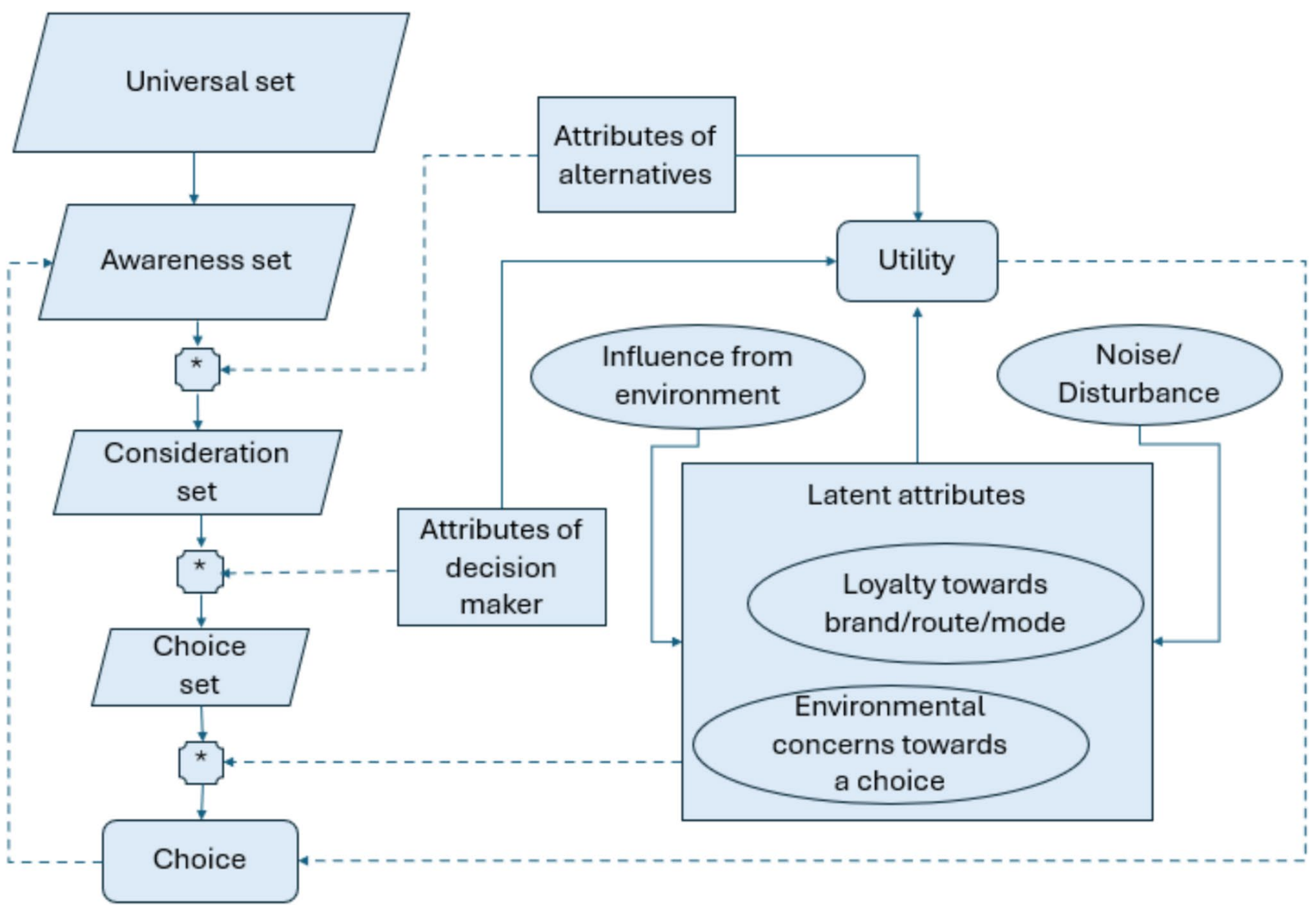

Fig. 1 Conceptual framework template

diverse agent behaviors and improve decision-making processes. While traditional set models are well-established and effective, the adoption of ML models enables the exploration of more complex relationships and patterns in data that may not be fully captured by traditional approaches. By comparing ML models with established choice set models, the study aims to evaluate the accuracy, robustness, and scalability of ML techniques in generating consideration sets. This approach seeks to validate the performance of ML models and demonstrate their potential to complement traditional methods, particularly in scenarios involving heterogeneous populations or large-scale datasets.

Assumptions and limitations that this work has revolved around the use of synthetic data, and the randomness introduced, along with the assumption that all agents have equal access to information and similar levels of awareness regarding the alternatives.

The main research questions that this work is attempting to answer are the following:

- Which are the most suitable methods via which consideration sets of decision-makers can accurately be predicted?
- What are the insights that can be gained by predicting considered alternatives?
- How well can predictive models behave when compared to existing set generation models?
- Is it feasible for set generation models to be integrated with predictive models to more accurately observe considered alternatives of decision-makers?

The remainder of this paper is structured as follows. "Literature Review" section reviews the relevant background and related work. "Methodology" section presents the methodological framework and problem formulations. "Application and Results" section applies the consideration set framework to a case study of route choice and discusses the results. "Conclusion" section concludes with discussion, limitations, and future research directions.

Literature Review

This section presents a brief overview of existing literature in the realm of choice and consideration set generation modeling.

Choice Set Generation Modeling

Choice set generation models identify the subset of options decision-makers consider from a larger set of available alternatives before making their final decision. This includes heuristic processes, probabilistic models, search models, among other methods (Richardson 1982). Several studies focus on the marketing realm (Swait and Erdem 2007; Barseghyan et al. 2021), as well as in transportation-related cases (Swait 1984). Various route choice set generation models have been proposed, addressing the impact of choice set on route choice (Cascetta et al. 1996; Ben-Akiva et al. 1999; Bekhor et al. 2006, 2008; Fosgerau et al. 2013). The Modified Logit Route Choice Model (C-Logit), proposed by Cascetta et al. (1996), addresses path overlapping issues in route choice models. Traditional logit models often assume independence among alternatives, leading to inaccurate probability estimates when alternatives share common attributes or paths (i.e., independence of irrelevant alternatives—IIA).

The Path-Size Logit (PSL) model by Ben-Akiva et al. (1999) extends the MNL model, to address overlapping paths in route choice modeling by introducing a path-size attribute to account for the correlation between alternative paths sharing common links. Ben-Akiva and Boccara (1995) studied the integration of stochastic constraints or elimination criteria and the impact of attitudes and perceptions on the choice set generation process into a unified modeling framework. Swait (2001) introduced a generalized extreme value (GEV) distributed discrete-choice model, known as Choice Set Generation Logit (GenL), to integrate choice set generation modeling into the specification process. The GenL model makes choice set creation explicitly dependent on decision-makers' preferences, incorporating covariate information to reflect external constraints.

Swait and Ben-Akiva (1987a) introduced the Parametrized Logit Captivity (PLC) probabilistic choice model, which generalizes the Dogit captivity model introduced by Gaudry and Dagenais (1979). The PLC model allows the IIA property to hold for some pairs of alternatives while maintaining distinctiveness for other alternatives. It posits that an individual is either captive to a single alternative or free to choose from a choice set, following an MNL model. The probability of captivity is expressed as a function of independent variables, allowing the PLC model to predict a more flexible range of responses to changes in these variables compared to Dogit or MNL models.

Consideration Set Formation

Consumers usually reduce their choices using heuristic decision processes (Hauser 2014). Insights from the evoked set/acceptable brands set/consideration set (Howard and Sheth 1969; Wright and Barbour 1977; Belonax 1979) can improve the accuracy of choice models (Horowitz and Louviere 1995). To address the complexity of these sets, various models have been proposed (Chiang et al. 1998), and their impact on consumer judgment and decision-making has been studied (Shocker et al. 1991). Pioneering brands often gain preference over follower brands, affecting the consideration process (Kardes et al. 1993). Screening rules in discrete-choice models revealed that options failing initial screenings are unlikely to be chosen, which helps in predicting consumer choices (Gilbride and Allenby 2004; Jedidi and Kohli 2005; Evgeniou et al. 2007; Hauser et al. 2010).

Consumers typically avoid less favorable options, which affect their choice probabilities and willingness to pay regarding the actual choice (Capurso et al. 2019, 2021). Different models for consideration set composition are used in marketing and brand evaluation (Shapiro et al. 1997; Roberts and Lattin 1991; Nedungadi 1990; Hauser and Wernerfelt 1990; Mehta et al. 2003; Paulssen and Bagozzi 2005; Roberts 1989). Consideration effects improve the explanatory power of empirical discrete-choice models, whether static (Roberts and Lattin 1991) or dynamic (Roberts and Nedungadi 1995). A rational consumer consideration set reflects a trade-off between decision-making costs and the marginal benefits of a larger set of options (Chakravarti and Janiszewski 2003).

Methodology

This section outlines the methodological framework introduced in this work. Existing choice set and consideration set generation models were used for constructing the consideration sets of the synthetic decision-makers. The considered ML models were then trained and tested on these sets. The training set consisted of consideration sets based on various routes agents might consider. Route consideration is determined by the attributes of each route and the agents' coefficients toward these attributes, reflecting their importance in the decision-making process.

A two-stage process was adopted, similar to previous route choice set generation modeling (Bovy 2009). To achieve practical specifications in most choice contexts, it is essential to limit the space of possible choice sets to a reasonable subset (Swait and Ben-Akiva 1987b).

Two approaches were studied. The first involved a homogeneous synthetic population in the sense that the same consideration set generation model was used by the whole population for consideration set construction. Multiple models were studied independently in this homogeneous case. The second approach involved a heterogeneous population, in which every agent was randomly assigned a consideration set generation model. Route attribute assignment was based on existing literature (Cascetta et al. 2002; Selten et al. 2007; Peeta and Yu 2006; Bovy and Bradley 1985; Papinski and Scott 2011).

For the homogeneous population case process (Fig. 2), a synthetic set of routes was generated, followed by an iterative process to create synthetic populations of agents with randomly assigned sociodemographic characteristics. In each iteration, the choice and consideration set generation models were applied to formulate each agent's consideration set. Subsequently, each agent's choice set was created by selecting the three options from their consideration set with the highest utility. The consideration and choice sets for all agents were stored in each iteration to form a superset representing "averaged" agents. This process mitigates the randomness introduced using synthetic data. Each ML predictive model was then trained and tested on the resulting datasets.

In the heterogeneous population case (Fig. 3), after route generation, a synthetic population of decision-makers was generated, with each agent getting randomly assigned one of the set generation models. After the consideration set was assembled, the ML predictive models were trained on this dataset. Following, a new (homogeneous) population of decision-makers was generated, with each agent's consideration set generated based on the same, logit-based set generation model. The ML predictive models, previously trained on the heterogeneous population, were then applied

Fig. 2 Homogeneous population case

Algorithm 1: Homogeneous population case flow of operations

```
Generate routes
While iteration ≤ max _iterations
    Generate synthetic population
        For each agent on population
            For each set generation model
                Generate consideration set
                Construct choice set
For each set generation model
    Construct averaged agent superset
    For each ML predictive model
        Train ML model
        For each agent on superset
            Predict choice set
Evaluate
Terminate
```

Fig. 3 Heterogeneous population case

Algorithm 2: Heterogeneous population case flow of operations

```
Generate routes
Generate synthetic population
    For each agent on population
        Assign set generation model
        Generate consideration set
        Construct choice set
For each ML predictive model
    Train ML model
Generate synthetic population
Assign set generation model
Generate consideration set
Construct choice set
For each agent on population
    Predict choice set
Evaluate
Terminate
```

to predict the consideration sets of everyone on the homogeneous population.

This approach allows for comparing the ML models with traditional models to evaluate the accuracy of ML predictions. The evaluation of the predictive capabilities of the ML models was conducted in both cases based on per-item comparison (i.e., considering routes individually as the response variable), and considering the whole set of alternative routes.

Consideration—Choice Set Generation Models

This section briefly presents the set generation models considered in the proposed framework.

Deterministic Utility

A deterministic utility choice set generation model is based on the premise that each decision-maker evaluates all available alternatives and selects a subset of these alternatives that maximizes their utility, under a deterministic utility function and perfect information. The utility function is given as $U_i = \beta X_i$, where X_i is a vector of attributes of alternative i from the awareness/universal set J. The choice set $C_n \subset J$ is generated by selecting alternatives that meet a minimum utility threshold, τ, where $C_n = \{i \in J | U_i > \tau\}$.

Random Utility

In random-utility models, it is assumed that a decision-maker's choice set is independent of their preferences, given the explanatory variables of the models, while in certain cases, the decision-maker selects the choice set, making the independence assumption unrealistic (Ben-Akiva and Boccara 1995; Manski 1997; Horowitz 1991; Walker and Ben-Akiva 2002). The utility (U_{in}) is composed of a deterministic component V_{in}, where $V_{in} = \beta x_{in}$, typically specified as a linear function of the attributes of the alternative x_{in} and parameters β, and a random error term ε_{in}, where $U_{in} = V_{in} + \varepsilon_{in}$.

Multinomial Logit (MNL)

Based on a comparable to the Multinomial Logit model introduced by Manski (1977), C_n is denoted as the choice set considered by an individual n, the choice model (Bierlaire et al. 2010) is the following (Eq. 1):

$$P\left(i|C_n\right) = \Pr\left(U_{in} \geq U_{jn}\right), \forall j \in C_n \tag{1}$$

where U_{in} corresponds to the random utility regarding alternative i, for individual n. If C_n is known, then Eq. 1 can be written as

$$P\left(i|C_n\right) = \Pr(U_{in} + \ln A_{in} \geq U_{jn} + \ln A_{jn}, \forall j \in C \tag{2}$$

where A_{in} is an indicator of consideration of each alternative, with A_{in} being equal to 1 if alternative i is considered by n, 0 otherwise (C the universal set). The probability of individual n choosing alternative i from choice set C_n is given by Eq. 3:

$$P(i) = \frac{e^{V_{in} + \ln A_{in}}}{\sum_{j \in C} e^{V_{jn} + \ln A_{jn}}}. \tag{3}$$

Latent Class

Latent Class models extend traditional discrete-choice models by accounting for unobserved heterogeneity in the population and integrating latent variables in decision models (Swait and Adamowicz 2001; Lapparent 2010). The population is divided into a finite number of latent classes, each with its own choice behavior and consideration set formation process. Within each class, individuals are assumed to have similar preferences and decision-making processes. The probability that individual n belongs to Latent Class q is given by $P(q|n) = \frac{e^{\gamma_q z_n}}{\sum_{q'} e^{\gamma_q z_n}}$, where z_n is a vector of observable characteristics of individual n, and γ_q a vector of parameters regarding class q. The probability that individual n i from the choice set C_n is given by Eq. 4:

$$P(i|n) = \sum_q P(q|n) \sum_{C_n \underline{C} C} P(C_n|q,n) P(i|C_n,q,n) \tag{4}$$

where the probability that individual n in class q generates a choice set C_n is denoted as $P(C_n|q,n) = \frac{e^{\alpha_q x_{C_n}}}{\sum_{C' \underline{C} C} e^{\alpha_q x_{C_n}}}$, where x_{C_n} is a vector of attributes of the choice set C_n, and a_q a vector of parameters specific to class q.

Modified Logit Route Choice (C-Logit)

The C-Logit model retains the structure of the MNL model, with a modified disutility (Eq. 5), where $\widehat{V_k^n} = V_k^n - CF_k$, where $V_k^n = \beta_i x_{ik}$, n individual decision-maker, $k \in K$ set of paths.

$$P(k|n) = \frac{e^{V_k^n - CF_k}}{\sum_{h \in K} e^{V_k^n - CF_k}}. \tag{5}$$

The term CF_k (Eq. 6) corresponds to a commonality factor and is directly proportional to the degree of overlapping of path k with other paths in set K. L_{hk} corresponds to the length of links common to paths h, k, and L_h and L_k the summations of link lengths of paths h and k, with $\gamma > 0$ model parameter. Therefore, paths with significant overlap have a higher CF_k, leading to a lower systematic utility with respect to similar but independent paths.

$$CF_k = \beta_0 \ln \sum_{h \in K} \left(\frac{L_{hk}}{L_h^{\frac{1}{2}} L_k^{\frac{1}{2}}} \right)^{\gamma}. \tag{6}$$

Generalized Logit (GenL)

The set of all possible subsets of M, where M is the master set of alternatives is denoted as $\Delta(M)$, and K the number of sets included in $\Delta(M)$, which is of size $1 \leq K \leq (2^J - 1)$, $M = 1, \ldots, J$. The GenL choice probability is given by Eq. 7:

$$P(i) = \sum_{k \underline{C} K_i} P(i, C_k) = \sum_{k \underline{C} K_i} P(i|C_k) Q(C_k) \tag{7}$$

where $P(k|n) = \frac{e^{\mu_k V_i}}{\sum_{j \in C_k} e^{\mu_k V_j}}, \forall i \in C_k, k = 1, \ldots, K$, $Q(C_k) = \frac{e^{\mu I_k}}{\sum_{r=1}^{K} e^{\mu I_r}}, k = 1, \ldots, K$, and $I_k = \left(\frac{1}{\mu_k}\right) \ln\left(\sum_{j \in C_k} e^{\mu_k V_j}\right), k = 1, \ldots, K$.

$$G(y_1, \ldots, y_J) = \sum_{k=1}^{K} \left(\sum_{j \in C_k} y_{jk}^{\mu} \right)^{\frac{\mu}{\mu_k}}, \mu \geq 0, k = 1, \ldots, K \tag{8}$$

Equation 8 presents the GEV generating function for the GenL model, where $G(y_1, \ldots, y_J)$ is a non-negative of degree μ homogeneous function.

Dogit (Logit Captivity)

The Dogit model assumes that a decision-maker can be either captive to an alternative within a full set of options known to that individual or is free to select among all alternatives according to a MNL model. The form of this model is shown in Eq. 9:

$$P_i = \frac{u_i}{1+u} + \left(\frac{1}{1+u}\right) P_{(i|C_n)}, \forall i \in C_n, \tag{9}$$

where u_i represents the probability that the individual is captive to $i \in C_n$, with $u = \sum_{j \in C_n} u_j$, and $P_{(i|C_n)}$ representing the MNL choice model. The term $\frac{u_i}{1+u}$ corresponds to the probability of captivity to $i \in C$, while $\left(1 + u^{-1}\right)$ is the probability of the decision-maker being able to choose from the full choice set C_n. Despite the MNL being included in the Dogit model, it was shown that it does not display the IIA property that the MNL does (Gaundry and Dagenais 1979).

Parametrized Logit Captivity (PLC)

While developing a version of the Dogit model, Ben-Akiva (1977) proposed that parameters u_i should be parametrized as functions of independent variables, meaning that since the u_i are bounded to be non-negative, leading to Eq. 10:

$$u(X_i) = e^{DX_i}, \forall i \in C_{n,} \tag{10}$$

where $D = (d_1, \ldots, d_k, \ldots, d_K)$ is a vector of parameters, and X_i is a vector of socioeconomic characteristics of the individual, and attributes of option i. Through Eqs. 9 and 10, the PLC model was derived (Eq. 11):

$$P_i = \frac{e^{DX_i}}{1 + \sum_{j \in C_n} e^{DX_j}} + \frac{1}{1 + \sum_{j \in C_n} e^{DX_j}} \left(\frac{e^{BY_i}}{\sum_{j \in C_n} e^{BY_i}} \right), \forall i \in C_n, \tag{11}$$

where $B = (b_1, \ldots, b_l, \ldots, b_L)$ is a vector of parameters of the MNL model, and Y_i is a vector of socioeconomic characteristics of the individual, and attributes of option i.

Path-Size Logit (PSL)

The PSL model applies the concepts of elemental alternatives and size variables, with overlapping paths may not be perceived as distinct alternatives since they share links, when applied in the context of route choice. Therefore, the size of a path with shared links may be less than one. To account for this, a size variable is included in the utility function of a path (Eq. 12).

$$P(i|C_n) = \frac{e^{\mu(V_{in}+lnS_{in})}}{\sum_{j\in C_n} e^{\mu(V_{in}+lnS_{jn})}} \tag{12}$$

The size S_{in} is defined as $S_{in} = \sum_{\alpha\in\Gamma_i} \frac{l_\alpha}{L_i}\left(\frac{1}{\sum_{j\in C_n}\delta_{\alpha j}\left(\frac{L^*_{C_n}}{L_j}\right)}\right)$, where Γ_i is the set of links in path i, l_α and L_i are the length of link α and path i, and $\delta_{\alpha j}$ is the link-path decision variable (equals to 1 if link α is on path j, 0 otherwise). $L^*_{C_n}$ is the length of the shortest path in the set C_n.

Machine Learning Consideration Set Predictive Models

This section briefly presents the predictive ML models considered in the proposed framework.

Binary Relevance Model

BR models handle multi-label classification problems by converting them into other learning instances, enabling the application of more than one specific learning method (Zhang et al. 2018). A BR model subdivides the multi-label learning task into q separate binary learning tasks. In the label space, each binary classification problem corresponds to one class label.

Classifier Chains Method

Classifier Chains models (CC) share some similarities with BR models, since binary classifiers are involved as well. The main difference is that they are linked in a chain, with each classifier handling a BR problem connected with each label (Read 2008; Read et al. 2011). Every classifier's (c_i) task is to first learn (via training) and then predict the binary association of the ith label, with the attribute space being increased by all the previous BR predictions that assemble the chain.

Decision Trees

The goal of Classification and Regression Tree methods is to achieve the most accurate prediction using a quantitative response variable and several predictors (Berk 2008; Breiman 2017). First, all feasible splits of the predictor values are explored for each predictor. The optimal split for each predictor is chosen based on minimizing the sum of squares of the response variable.

Second, after determining the optimal split for each predictor, the best overall split is identified using the same sum of squares criterion. The best predictor is implicitly selected by finding the split that reduces the sum of squares the most. This optimal split is then used to divide the population into two groups, effectively sub-setting the data based on the best predictor's split.

Random Forest

A Random Forest (RF) model is an ensemble classification model comprising many Decision Trees, created through bagging and random variable selection (Wu et al. 2019). Trees in an RF are built using recursive partitioning, where the selection of cut-points and splitting variables depends on the distribution of observations in the learning sample. This makes trees unpredictable, as small changes in the data can alter the tree's structure. However, RF addresses this by combining multiple Decision Trees. The RF classifier consists of several tree classifiers, each built using a random vector that is uncorrelated with the input vector. Each tree votes for the most popular class to classify the given data (Breiman 2001).

Label Powerset

The Label Powerset (LP) model exploits correlations between labels, and for each alternative combination of labels, it creates a single-label dataset with a separate class (Tsoumakas et al. 2010; Moyano et al. 2018). LP incorporates label correlations, but its complexity grows exponentially with the number of labels, and it cannot predict a labelset that isn't present in the training dataset.

XGboost

The Gradient Boosting algorithm was designed for predictive operations and is based on building one Decision Tree at a time to reduce errors from all preceding trees in the model (Friedman 2001). An adaptation of this algorithm was later introduced, called XGboost (Chen and Guestrin 2016), with the difference that Gradient Boosting was done by utilizing multiple cores to construct individual trees and organizing data to minimize lookup times.

Support Vector Machine

A Support Vector Machine model (Cortes 1995) is based on a learning algorithm designed for two-class classification tasks. It operates by non-linearly mapping input vectors into a high-dimensional feature space, where a linear decision boundary is then established. The unique characteristics of this decision boundary enable the learning machine to achieve strong generalization capabilities. This concept can be applied to cases where training data could be perfectly separated, as well as scenarios involving non-separable training data.

Assessment Measures

A wide range of measures has been developed to evaluate multi-label classification methods (Tsoumakas and Katakis 2008). In this study, a selected subset of these measures was employed. Let S denote a multi-label dataset consisting of $|S|$ instances (x_i, y_i), where $i = 1, \ldots, |S|$, and y_i is a subset of the total number of labels $|N|$. Let C be a multi-label classifier, with $R_i = C(x_i)$ representing the set of labels predicted by C for instance x_i. Among the more intuitive and commonly evaluation metrics are accuracy, precision, and recall, (Godbole and Sarawagi 2004), which are defined in Eqs. 13–15:

$$\text{Accuracy}(C, S) = \frac{1}{|S|} \sum_{i=1}^{|S|} \frac{|y_i \cap R_i|}{|y_i \cup R_i|} \tag{13}$$

$$\text{Precision}(C, S) = \frac{1}{|S|} \sum_{i=1}^{|S|} \frac{|y_i \cap R_i|}{|R_i|} \tag{14}$$

$$\text{Recall}(C, S) = \frac{1}{|S|} \sum_{i=1}^{|S|} \frac{|y_i \cap R_i|}{|y_i|}. \tag{15}$$

Accuracy refers to the proportion of correctly predicted labels relative to the total number of instances. Precision measures the fraction of relevant labels among all labels predicted for a given class. An alternative definition of precision considers the percentage of true positives among all positive predictions. Recall quantifies the proportion of actual labels that were correctly predicted by the model; it can also be expressed as the ratio of true positives to the total number of relevant instances.

In addition to these metrics, Hamming loss (Schapire and Singer 2000) captures both types of prediction errors: prediction error (when an inaccurate label is predicted) and missing error (when a valid label is not considered). These errors are standardized over the total number of labels and instances, as shown in Eq. 16:

$$\text{Hamming\_Loss}(C, S) = \frac{1}{|S|} \sum_{i=1}^{|S|} \frac{|y_i \Delta R_i|}{|N|} \tag{16}$$

where Δ denotes the symmetric difference between two sets.

Application and Results

This section presents the numerical analysis followed to incorporate the previously introduced framework.

Data

Synthetic data were used, for the route generation, as well as the decision-makers, with randomly assigned attributes, sampled from 2015 Census and American Community Survey (ACS) data, obtained from the Chicago Metropolitan Area Planning (CMAP) (2018). Route attributes such as cost, travel time, congestion probability, and accident rates were sampled from uniform or normal distributions within specified ranges, with mean and standard deviation values derived to mimic realistic variability. Categorical attributes like road type and toll presence were sampled from discrete probability distributions to represent their typical occurrences. Continuous variables like age, income, and household size were sampled from normal distributions with mean and standard deviation values that align with the demographic and economic profile of the region. Categorical variables, including gender, employment status, education level, ethnicity, and political views, were assigned probabilities based on their observed proportions in the CMAP data. Table 1 describes the data considered.

Numerical Analysis

Different scenarios were considered for the two cases, with accuracy (number of correct predictions divided by the total number of instances), precision (percentage of real positives among all positive predictions), recall (proportion of instances of a given class that the model correctly predicted as belonging to that class), Hamming loss (prediction error (when an inaccurate label is predicted), and missing error (when a valid label is not considered), standardized over the total number of classes and instances) being the assessment metrics considered. Figures 4 and 5 present a general overview of the results, with Fig. 4 showing different variations in the accuracy levels of each ML model considered, varied by the different set generation methods. From a set generation standpoint, the MNL and the random-utility models showed generally the highest accuracy levels when utilized with the ML models. In addition, the C-Logit model also showed significant accuracy scores, especially when utilized by the RF, XGBoost, and CC algorithms. This could be the case because the MNL and random-utility models rely on probabilistic frameworks that closely resemble the structure of many ML classification models, making them more compatible for training predictive models. Their ability to

Table 1 Route and decision-maker attributes

Attribute	Range/categories	Mean ± SD
Route set		
Cost ($)	[10–100]	45 ± 23
Travel time (minutes)	[30–240]	93 ± 52.5
Tolls ($)	0 (no toll) or 1 (toll)	–
Congestion (probability)	[0–1]	0.5 ± 0.29
Scenery (rating)	{1–5}	–
Safety (rating)	{1–5}	–
Reliability (rating)	{1–5}	–
Comfort (rating)	{1–5}	–
Road type	1 (highway), 2 (urban street), 3 (rural road)	–
Accident rate (probability)	[0–1]	0.2 ± 0.14
Decision-maker set		
Age	{18–75}	41 ± 17.5
Gender	Male, female, other	–
Income	[20,000–250,000]	105,000 ± 57,735
Household size	1, 2, …, 5 or more	3 ± 1.41
Driver's license	0 (no) or 1 (yes)	–
Employment status	Employed, unemployed, Student, retired, part-time	–
Education	Middle school, high school, technical school, college, Graduate degree	–
Vehicles on household	0, …, 4 or more	–
Daily work trips	{0,3}	1.5 ± 1.42
Daily non-work trips	{0,7}	2.5 ± 1.74
Ethnicity	Hispanic, African American, White, Asian, Hispanic, Native American, Other	–
Typical mode of transportation	Personal vehicle, public transit, bike, walking, other	–
Political views	Liberal, conservative, moderate, independent, very conservative, very liberal	–

Fig. 4 Accuracy scores for considered ML models and set generation methods

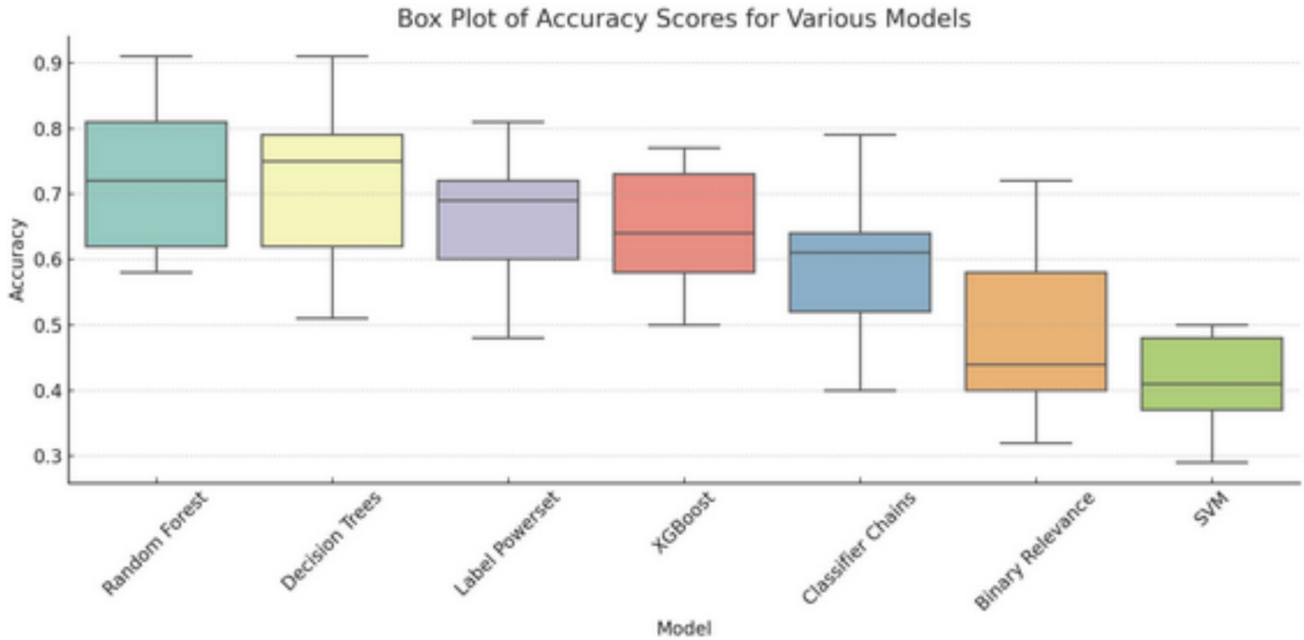

Fig. 5 Accuracy scores of ML models

capture varying sensitivities to route attributes allows ML models to learn more distinct patterns, leading to higher predictive accuracy. Similarly, the C-Logit model, which accounts for path overlap effects, may provide a more refined consideration set that aligns well with the feature importance mechanisms of ensemble methods like RF and XGBoost.

From the ML models perspective, based on Figs. 4 and 5, RF was found to provide higher accuracy scores, along with Decision Trees, with the former, however, showing a smaller deviation in the provided scores, along with a higher lower bound of accuracy than the former. In addition, LP and CC were able to provide encouraging scores up to 80%. This suggests that ensemble-based models like RF benefit from their ability to reduce variance and overfitting by aggregating multiple Decision Trees, leading to more stable and consistently high accuracy scores. The smaller deviation in RF's accuracy further indicates its robustness across different set generation methods, making it a reliable choice for consideration set prediction. Decision Trees, while also performing well, exhibit a wider range of accuracy scores, likely due to their sensitivity to variations in training data and a lack of ensemble smoothing. In addition, the strong performance of LP and CC highlights the effectiveness of multi-label classification techniques in capturing the interdependencies between considered alternatives, allowing for more nuanced and context-aware predictions.

Table 2 presents the results of considered ML models for the homogeneous case, trained and tested per item/route (average metrics per route) on the entire response set, and on populations with consideration sets generated via the considered set generation models. The sample size was 4000 individuals, with 6 alternatives/routes, allowing for up to 4 of them to be included in their consideration set. It can be observed that RF, followed by XGBoost, LP and Decision Trees, were the models presenting the higher accuracy, with the MNL and C-Logit being among the set generation models leading to prediction accuracy between 75 and 85%, per-item prediction, and satisfying recall, precision, and relatively low Hamming loss values. In the entire set prediction case, Decision Trees showed higher accuracy results, though lower than the per-item case as expected, but still at satisfying levels (between 60 and 68%, depending on the set generation model considered, with the MNL and C-Logit again having the highest accuracy levels).

In the heterogeneous case (Table 3), RF again offered the highest accuracy. The per-item prediction accuracy was lower compared to the homogeneous case but still above the threshold of 70%. In the entire set prediction case, the MNL and C-Logit models combined with RF continued to present the highest accuracy levels.

The performance differences observed across the set generation models in Table 2 are closely related to the structural assumptions each model imposes on how consideration sets are formed, and how these assumptions interact with the ML predictive layer. In the per-item prediction setting, models such as MNL and GenL tend to achieve higher predictive accuracy because they generate consideration sets that are more systematic, stable, and closely aligned with the attributes of the alternatives. This results in training data that is less noisy and more conductive to learning by ML

Table 2 Homogeneous case results

	Deterministic utility	Random utility	MNL	Latent Class	GenL	Dogit	C-Logit	PLC	PSL
	*Per item prediction (**accuracy**, recall, precision, Hamming loss)*								
Random Forest	**0.78**	**0.72**	**0.84**	**0.61**	**0.71**	**0.72**	**0.81**	**0.58**	**0.62**
	0.88	0.69	0.73	0.52	0.48	0.56	0.71	0.32	0.47
	0.91	0.71	0.79	0.69	0.61	0.78	0.85	0.41	0.51
	0.16	0.15	0.11	0.13	0.18	0.21	0.12	0.31	0.39
Decision Trees	**0.75**	**0.61**	**0.71**	**0.52**	**0.79**	**0.66**	**0.62**	**0.51**	**0.76**
	0.91	0.72	0.67	0.56	0.61	0.71	0.51	0.41	0.65
	0.84	0.70	0.69	0.61	0.54	0.65	0.60	0.38	0.58
	0.05	0.12	0.11	0.16	0.14	0.20	0.21	0.29	0.21
Label Powerset	**0.69**	**0.68**	**0.80**	**0.60**	**0.78**	**0.72**	**0.69**	**0.48**	**0.55**
	0.68	0.67	0.62	0.43	0.45	0.49	0.33	0.30	0.41
	0.81	0.66	0.77	0.69	0.52	0.71	0.65	0.33	0.48
	0.08	0.08	0.11	0.13	0.19	0.24	0.25	0.31	0.38
XGBoost	**0.72**	**0.65**	**0.77**	**0.50**	**0.73**	**0.63**	**0.76**	**0.54**	**0.58**
	0.64	0.69	0.65	0.42	0.55	0.42	0.74	0.45	0.67
	0.77	0.71	0.72	0.57	0.44	0.64	0.69	0.29	0.39
	0.12	0.09	0.09	0.11	0.16	0.21	0.17	0.27	0.29
Classifier Chains	**0.63**	**0.62**	**0.59**	**0.53**	**0.61**	**0.48**	**0.64**	**0.40**	**0.52**
	0.79	0.63	0.54	0.39	0.37	0.44	0.29	0.28	0.32
	0.73	0.51	0.76	0.58	0.54	0.64	0.59	0.32	0.44
	0.03	0.11	0.13	0.12	0.22	0.26	0.26	0.32	0.39
Binary elevance	**0.45**	**0.54**	**0.50**	**0.32**	**0.58**	**0.43**	**0.44**	**0.40**	**0.40**
	0.59	0.35	0.46	0.37	0.31	0.35	0.17	0.25	0.33
	0.72	0.31	0.56	0.41	0.45	0.37	0.58	0.19	0.24
	0.11	0.10	0.13	0.18	0.22	0.27	0.28	0.37	0.47
SVM	**0.39**	**0.50**	**0.39**	**0.41**	**0.50**	**0.49**	**0.33**	**0.37**	**0.29**
	0.48	0.41	0.42	0.49	0.29	0.31	0.29	0.31	0.27
	0.45	0.36	0.44	0.47	0.50	0.42	0.45	0.16	0.31
	0.25	0.19	0.21	0.22	0.29	0.32	0.31	0.39	0.44
	*Entire set prediction (**accuracy**, recall, precision, Hamming loss)*								
Random Forest	**0.42**	**0.49**	**0.62**	**0.11**	**0.15**	**0.13**	**0.61**	**0.16**	**0.12**
	0.56	0.51	0.62	0.09	0.14	0.21	0.43	0.15	0.15
	0.50	0.39	0.46	0.10	0.12	0.24	0.49	0.21	0.12
	0.33	0.36	0.32	0.51	0.54	0.46	0.35	0.31	0.62
Decision Trees	**0.47**	**0.49**	**0.68**	**0.17**	**0.18**	**0.24**	**0.65**	**0.11**	**0.13**
	0.60	0.61	0.51	0.31	0.24	0.20	0.54	0.13	0.18
	0.51	0.42	0.50	0.18	0.14	0.18	0.59	0.18	0.21
	0.21	0.30	0.15	0.46	0.40	0.44	0.21	0.52	0.56
Label Powerset	**0.29**	**0.32**	**0.39**	**0.09**	**0.10**	**0.15**	**0.32**	**0.12**	**0.10**
	0.41	0.43	0.51	0.12	0.11	0.21	0.34	0.09	0.11
	0.43	0.30	0.56	0.17	0.17	0.20	0.37	0.07	0.21
	0.32	0.38	0.30	0.58	0.54	0.52	0.32	0.58	0.59
XGBoost	**0.44**	**0.51**	**0.60**	**0.10**	**0.13**	**0.16**	**0.59**	**0.12**	**0.15**
	0.61	0.49	0.55	0.06	0.18	0.25	0.51	0.12	0.17
	0.49	0.41	0.41	0.14	0.09	0.28	0.45	0.18	0.16
	0.29	0.32	0.37	0.57	0.56	0.44	0.43	0.27	0.54

Table 2 (continued)

	Deterministic utility	Random utility	MNL	Latent Class	GenL	Dogit	C-Logit	PLC	PSL
Classifier Chains	**0.24**	**0.26**	**0.21**	**0.08**	**0.07**	**0.11**	**0.08**	**0.07**	**0.08**
	0.26	0.21	0.34	0.11	0.09	0.22	0.12	0.11	0.12
	0.31	0.32	0.24	0.14	0.11	0.18	0.16	0.09	0.15
	0.41	0.44	0.45	0.56	0.62	0.59	0.61	0.65	0.64
Binary Relevance	**0.19**	**0.12**	**0.14**	**0.06**	**0.06**	**0.08**	**0.06**	**0.05**	**0.05**
	0.13	0.12	0.20	0.09	0.08	0.11	0.05	0.06	0.07
	0.16	0.07	0.15	0.08	0.12	0.19	0.08	0.10	0.12
	0.56	0.62	0.52	0.67	0.69	0.68	0.71	0.70	0.68
SVM	**0.16**	**0.34**	**0.19**	**0.07**	**0.06**	**0.08**	**0.06**	**0.08**	**0.06**
	0.21	0.35	0.29	0.12	0.11	0.19	0.14	0.09	0.11
	0.29	0.24	0.27	0.17	0.15	0.14	0.15	0.13	0.18
	0.49	0.53	0.49	0.52	0.58	0.54	0.60	0.66	0.61

Bold values highlight accuracy among four different metrics for each model across different utility types

algorithms like RF and XGBoost, leading to consistently higher precision and lower Hamming loss. By contrast, models like Latent Class and PLC introduce greater heterogeneity or stochasticity in set construction. While this better reflects realistic behavioral variation, it also increases variance in the training labels, making it more difficult for ML classifiers to detect stable patterns in per-item prediction. Consequently, the superior performance of MNL and GenL in per-item results does not necessarily imply that they are better behavioral models overall, but rather that they produce training outputs that are more compatible with the discriminative structure of ML algorithms.

In the entire set prediction case, the evaluation metric is more demanding, since the model must correctly identify the exact combination of considered alternatives rather than access each item's inclusion independently. Under this framework, the impact of structural assumptions is amplified. Although Latent Class and GenL models provide relatively high per-item accuracy, they tend to produce larger variability in the combinations of sets across individuals. This variability poses a challenge for ML models, making it significantly more difficult to reconstruct the exact sets. This explains the significant drop in their performance compared to models like MNL or C-Logit, which generate more consistent sets with clearer attribute-driven inclusion rules. The shift in relative performance from per-item to entire set prediction does not imply that one family of models is inherently "better" or "worse," but rather highlights the sensitivity of ML predictive success to the degree of stability, complexity, and variance embedded in the generated sets.

In summary, the differences across models reflect two key mechanisms: (1) the degree to which each set generation model's outputs align with the learning assumptions of ML models in per-item prediction, and (2) the extent of variability and combinatorial complexity they introduce in entire set prediction. Models like MNL and C-Logit perform more strongly not because they dominate behaviorally, but because their outputs are more compatible with the supervised learning paradigm. In contrast, models such as Latent Class and GenL, while behaviorally richer, introduce greater variability that challenges the ML layer—particularly when the task shifts from predicting individual item inclusion to reconstructing full consideration sets—ultimately reducing predictive accuracy.

Sensitivity Analysis

Figures 6 and 7 present the first sensitivity analysis for the homogeneous and the heterogeneous cases, respectively, examining how three key parameters, the number of alternatives (routes), consideration set size, and decision–maker population size, affect prediction accuracy. In both scenarios, RF was used as the predictive model, while the MNL model served as the set generation mechanism for the homogeneous population and the homogeneous subset of the heterogeneous case. Reference thresholds of 0.6 (60%) and 0.7 (70%) were established based on existing multi-label classification literature and calibrated using synthetic data experiments, given the variability in datasets and problem settings across prior studies. Accuracy values ranged from 0.3 to 0.88 (Sucar et al. 2014; El-Hasnony et al. 2022; Sun et al. 2017) across both analyses.

In the homogeneous case (Fig. 6), prediction accuracy exhibits systematic variation with model parameters. Consideration set size and the number of routes were the primary influencers, while population size show comparatively little effect. Specifically, smaller consideration sets generated higher accuracy, reaching up to 0.855 with 20 routes,

Table 3 Heterogeneous case results

	MNL	GenL	Dogit	C-Logit	PLC	PSL
	*Per item prediction (**accuracy**, recall, precision, Hamming loss)*					
Random Forest	**0.74**	**0.54**	**0.59**	**0.76**	**0.66**	**0.64**
	0.50	0.35	0.61	0.67	0.56	0.59
	0.51	0.55	0.66	0.59	0.71	0.45
	0.16	0.32	0.25	0.14	0.22	0.24
Decision Trees	**0.65**	**0.49**	**0.52**	**0.68**	**0.54**	**0.58**
	0.51	0.43	0.45	0.61	0.43	0.52
	0.48	0.44	0.55	0.56	0.51	0.56
	0.32	0.36	0.31	0.25	0.30	0.35
XGBoost	**0.69**	**0.54**	**0.56**	**0.72**	**0.62**	**0.62**
	0.56	0.49	0.39	0.68	0.56	0.62
	0.45	0.47	0.49	0.59	0.68	0.53
	0.29	0.32	0.35	0.22	0.26	0.32
Label Powerset	**0.69**	**0.53**	**0.48**	**0.72**	**0.49**	**0.44**
	0.62	0.38	0.45	0.56	0.51	0.41
	0.69	0.41	0.44	0.71	0.35	0.51
	0.21	0.22	0.29	0.15	0.31	0.33
	*Entire set prediction (**accuracy**, recall, precision, Hamming loss)*					
Random Forest	**0.64**	**0.38**	**0.54**	**0.61**	**0.42**	**0.40**
	0.57	0.51	0.57	0.69	0.46	0.38
	0.50	0.39	0.54	0.57	0.32	0.22
	0.17	0.36	0.28	0.21	0.34	0.37
Decision Trees	**0.58**	**0.42**	**0.40**	**0.55**	**0.39**	**0.37**
	0.38	0.32	0.42	0.54	0.32	0.42
	0.54	0.29	0.32	0.38	0.29	0.45
	0.24	0.34	0.41	0.29	0.33	0.38
XGBoost	**0.62**	**0.41**	**0.49**	**0.59**	**0.37**	**0.35**
	0.43	0.30	0.43	0.51	0.39	0.39
	0.58	0.26	0.29	0.33	0.41	0.48
	0.21	0.32	0.37	0.31	0.37	0.32
Label Powerset	**0.52**	**0.38**	**0.46**	**0.48**	**0.34**	**0.31**
	0.45	0.22	0.29	0.45	0.25	0.22
	0.49	0.25	0.24	0.49	0.23	0.34
	0.27	0.41	0.31	0.29	0.35	0.37

Bold values highlight accuracy among four different metrics for each model across different utility types

whereas larger consideration sets correspond to reduced accuracy, with values as low as 0.580. Figure 6d shows a positive correlation between the number of routes and accuracy, Fig. 6e shows a negative correlation with consideration set size, and Fig. 6f shows limited correlation with population size. Distribution plots (Fig. 6g–i) indicate that larger route sets are associated with higher mean accuracy but greater variability, while only small consideration sets consistently exceed the 70% threshold. Overall, accuracy ranges from below 0.46 to a peak of 0.858, with optimal performance observed in scenarios combining high route counts with small consideration sets.

In contrast, the heterogeneous case (Fig. 7) generates lower and more concentrated accuracy values, ranging from 0.378 to 0.613. The relationships between parameters and accuracy differ from the homogeneous case: Fig. 7a reveals a negative correlation between the number of routes and accuracy, with 4-route configurations achieving up to 0.613, while 20-route scenarios fell below 0.431. Figure 7b suggests that consideration set size has little systematic influence, while Fig. 7c shows negligible effects of population size. Distributional results (Fig. 7g–i) highlight increased variability across all groups, with no configuration consistently exceeding 60% accuracy and median values around 0.54. The contour and trend plots (Fig. 7d–f) reinforce the decline in accuracy with increasing route numbers. Across all configurations, accuracy

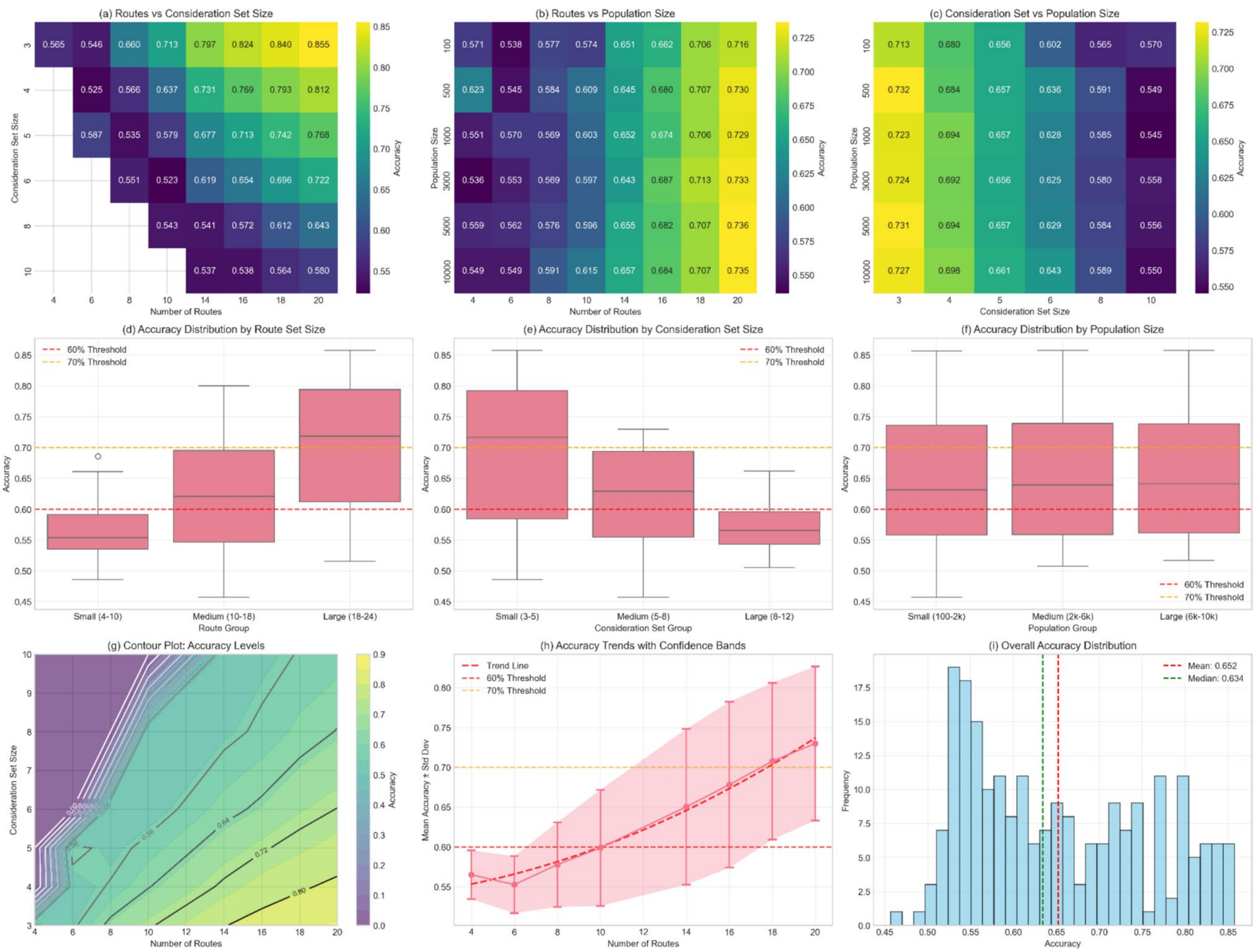

Fig. 6 Homogeneous case: Entire set prediction

ranges from 0.099 to 0.651, with the upper bound remaining below that of the homogeneous case (0.858).

The second sensitivity analysis that was done accounted for overlap between routes, to study its potential impact in the predictive capabilities of the models. Overlap between routes was modeled in a similar way to how Cascetta et al. (1996) approached it (similar way in principle, however, the mechanics in this study are quite simplistic due to the nature of the dataset used). A similarity index between two alternative routes was formulated (Eq. 17), grounded in the principles of the Paired Combinatorial Logit model, initially proposed by Chu (1989) and subsequently adapted by Koppelman and Wen (2000). In this set of routes Δ, each route δ is discretized into distinct links l, that assemble a given route. The travel time of each link is calculated based on a combination of free-flow average speed and additional delays caused by congestion and accidents. Congestion delays are probabilistically assigned based on the congestion probability of the given route, while accident delays are assumed to follow a uniform distribution across the links of the route. The similarity index is derived by considering the total travel time of links of road type n common to routes δ and ρ ($lt^n_{\delta\rho}$), and the total route travel time of road type n routes δ and ρ ($rt^n_\delta, rt^n_\rho \in \Delta^\delta, \Delta^\rho$, where $\Delta^\delta, \Delta^\rho \subseteq \Delta$).

$$o_\delta = -\frac{\sum_{\rho \in \Delta} lt^n_{\delta\rho}}{\sqrt{\sum_{\delta \in \Delta^\delta} rt^n_\delta + \sum_{\rho \in \Delta^\rho} rt^n_\rho}} \tag{17}$$

The term o_δ corresponds to the systematic disutility that route δ has to a given individual due to its potential overlap with other routes.

The second sensitivity analysis, presented in Figs. 8 and 9, explores the impact of various factors on prediction accuracy when considering overlapping between routes and using the Modified Logit Route Choice Model (Cascetta et al. 1996), instead of the MNL model. Similar to the previous one, this analysis examines the effect of the size of the universal set (number of alternatives, Figs. 8a, 9a), the size of the consideration set (Figs. 8b, 9b), and the decision-maker

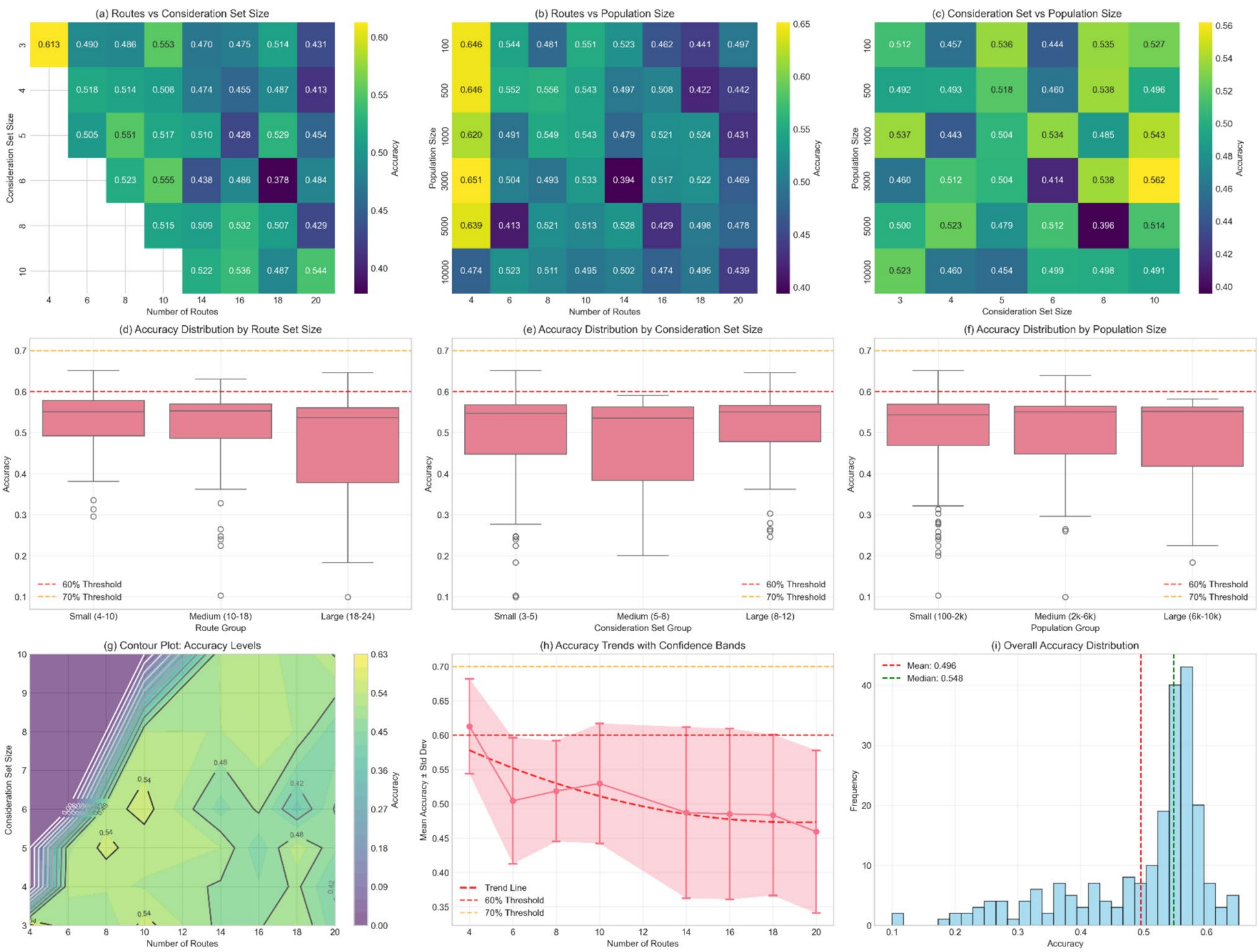

Fig. 7 Heterogeneous case: Entire set prediction

population size (Figs. 8c, 9c). Random Forest served as the predictive model, using the entire set for prediction.

For the homogeneous case, prediction accuracy is primarily influenced by the number of alternatives, while sample size and consideration set size have a more modest impact. Accuracy tends to improve as the number of alternatives increases beyond certain thresholds, with observed values ranging from approximately 50% to 83%. When examining homogeneous populations with route overlap, more nuanced patterns emerge. A strong non-linear relationship with the number of routes is observed: small route sets (4 routes) achieve peak accuracy (around 0.812–0.834), intermediate sets (8–10 routes) experience substantial declines (0.504–0.515), and larger sets (up to 20 routes) show a steady recovery (0.724). In this context, consideration set size and population size have minimal influence on performance, indicating that route-related dynamics are the dominant factor in shaping the impact of overlap. Distributional analysis shows a bimodal pattern: small route sets exhibit high variability, medium-sized sets consistently underperform, and large sets offer moderate but reliable accuracy.

In the heterogeneous case, prediction accuracy is more sensitive to variability among decision-makers, with sample size, consideration set size, and the number of alternatives all contributing to fluctuations. In the standard heterogeneous scenario, observed accuracy ranges from roughly 10% to 70%. When overlap is introduced, the prediction landscape becomes more challenging: accuracy is compressed into a narrower range (0.471 to 0.650), and correlations with key parameters are weak (routes: − 0.108; consideration set size: − 0.005; population size: 0.062). Overall performance remains generally low and highly variable, indicating that overlap effects create interference patterns that reduce the reliability of traditional optimization strategies. Although the best observed configuration achieves 0.736 accuracy, performance across scenarios remains unpredictable.

These results highlight that overlap introduces scenario-specific challenges. In homogeneous populations, route selection requires careful balancing: small sets can achieve

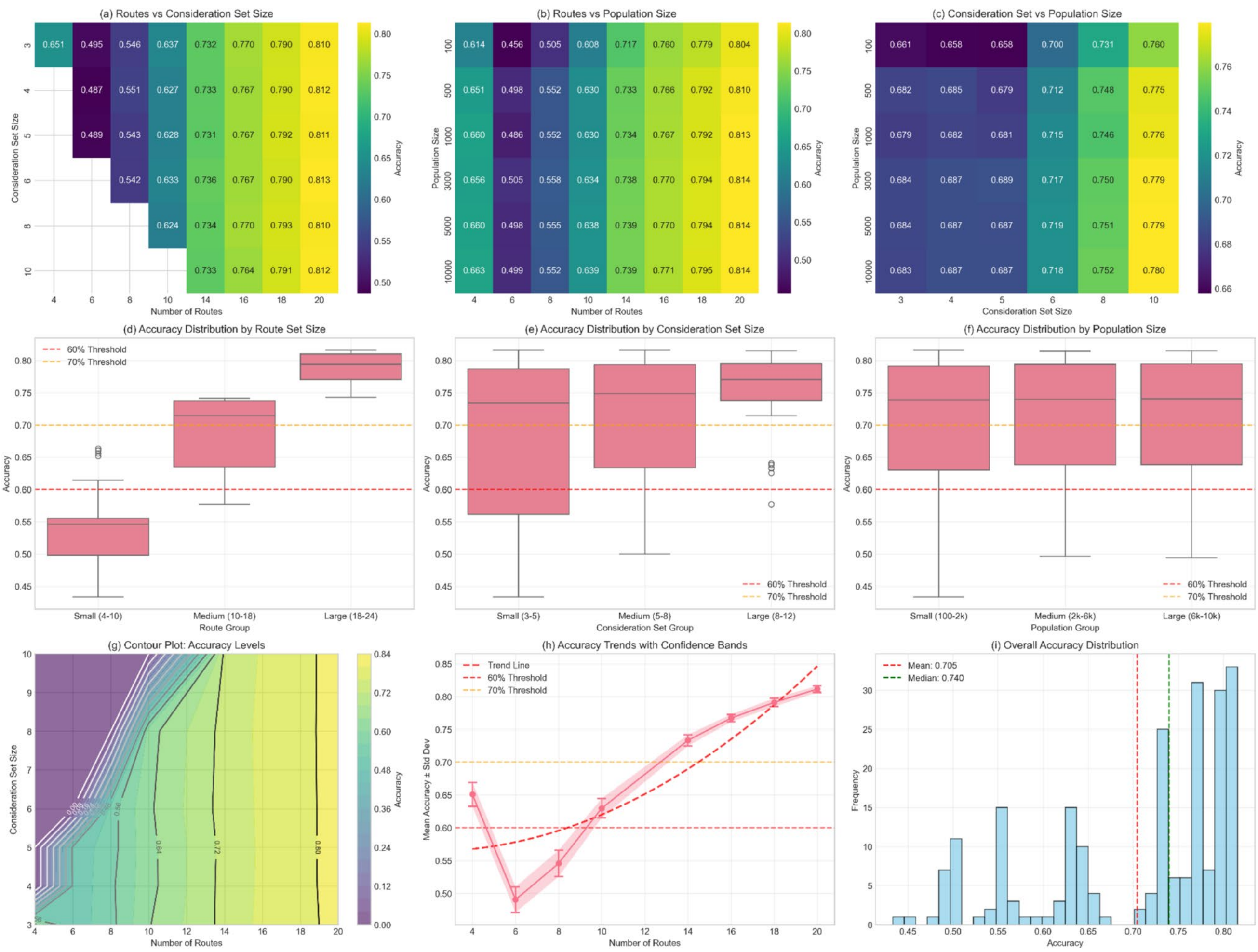

Fig. 8 Homogeneous case: Entire set prediction (overlapping routes)

peak performance but with high variability, while larger sets provide more consistent, moderate accuracy. Intermediate sets (e.g., 8–12 routes) tend to be less reliable and may be avoided. In heterogeneous populations, traditional parameter tuning proves less effective, indicating that alternative modeling approaches may be necessary to address the complexity introduced by overlapping preferences.

To summarize, the homogeneous analysis shows strong parameter interactions, with accuracy primarily driven by consideration set size and the number of routes. Small consideration sets consistently yield the highest performance, reaching up to 0.855 with 20 routes, whereas larger sets sharply reduce accuracy to around 0.580. Correlation results confirm this pattern, with routes positively linked to accuracy (0.570) and consideration set size negatively linked (− 0.536), while population size has little effect. Distribution analysis reinforces these findings: larger route sets increase mean accuracy but also variability, while small consideration sets provide the only consistent pathway to surpassing 70% accuracy. Population size shows negligible influence across all groups. Overall, the best-performing configuration (20 routes, consideration set 3, population 10 k) achieves 0.858 accuracy, while poor choices fall below 0.46. These results highlight that maximizing routes and minimizing consideration set size are key to strong performance, with population size only a secondary concern.

In contrast, the heterogeneous case reveals a very different landscape with lower and more compressed accuracies (0.378–0.613) and reversed parameter effects. Here, the number of routes correlates negatively with accuracy (− 0.219), as 4-route configurations achieve the best outcomes (0.613) while increasing to 20 routes reduces performance to below 0.431. Consideration set size no longer has a clear effect, and population size remains negligible. Distribution analysis further shows that small route sets outperform larger ones, though variability is much higher than in the homogeneous case, with no group consistently reaching 60% accuracy. The performance space is fragmented, with no clear optimal region: accuracy declines as routes increase, and uncertainty grows with larger configurations.

Fig. 9 Heterogeneous case: Entire set prediction (overlapping routes)

The best setup achieves only 0.651 accuracy, while the worst collapses to 0.099. These findings suggest that in heterogeneous populations, smaller, simpler configurations (4–6 routes, moderate sets) are preferable, though overall performance is inherently lower and less predictable compared to the homogeneous case.

Conclusion

This study introduces a comparative framework between consideration and choice set generation models and predictive models, focusing on the significance of the set of alternatives considered during the decision-making process before the final choice. The framework was applied to a synthetic route choice problem, where routes with randomly assigned attributes and synthetic populations of decision-makers with randomly assigned socioeconomic data were considered. Subsequently, ML models were trained and tested on these consideration sets. Two main approaches were employed: a homogeneous population-based approach, where the same set generation model was assigned to the entire population, and a heterogeneous population-based approach, where each decision-maker was randomly assigned a different set generation model. The primary goal was to assess the predictive accuracy of various ML models in predicting the alternatives considered by decision-makers and to determine if these models can be compared with and used analogously to existing set generation models.

Promising results were observed, demonstrating that ML models can achieve accuracy beyond 80% in per-item predictions and beyond 60% when considering the whole set of alternatives as the response variable. The heterogeneous case was found to have lower average accuracy levels than the homogeneous case, remaining, however, above the threshold of 60% and 70% accuracy for the cases of entire set prediction and per-item prediction, respectively. This relatively high level of predictive accuracy of ML models with the integration of specific set generation models underscores the effectiveness of ML models in capturing difficult-to-observe aspects of the decision-making process and can potentially lead to more accurate choice model estimations. In addition, the study emphasizes the flexibility of the proposed framework, which, despite its reliance on synthetic data, provides a generalizable testing environment for various decision-making contexts. Ultimately, the results underscore the potential of integrating ML techniques into consideration set modeling, offering a scalable and data-driven approach to understanding decision-making behavior.

The limitations in this work stem from the reliance on solely synthetic data, and the randomness that comes with such an approach. However, the ability to have a general scenario testing environment based on the way that the framework is defined provides the analyst with a flexible framework that can be applied to different problems.

In addition, while this study demonstrates that ML models can outperform traditional discrete-choice models in terms of predictive accuracy, it is important to acknowledge the challenge of interpretability often associated with ML approaches. Unlike discrete-choice models, which are structured and parameterized to provide clear insights into decision-making processes, ML models can be less easy to interpret and understand the mechanism behind these models. To address this limitation, tools such as feature importance analysis, Shapley Additive Explanations (SHAP), and Local Interpretable Model-agnostic Explanations (LIME) can be employed to explain individual predictions and identify key predictors driving the models' outputs (Lundberg 2017; Ribeiro et al. 2016). These techniques could offer a way to interpret the influence of socioeconomic and route attributes on predicted consideration sets. However, as pointed by Ruding et al. (2022), techniques like SHAP and LIME are designed to explain black box models but are unnecessary for inherently interpretable models. These methods quantify the contribution of each variable to a prediction. However, interpretable models inherently display the variables they use and how they are applied. In addition, ML models should be viewed as complementary to traditional models, leveraging their superior predictive capabilities while maintaining the insights provided by conventional methods.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflicts of interest The authors declare no competing interests.

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